TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 20

5-MINUTE REVISION

Single Factor Experiments

From the chapter's own revision points, formulae and questions.

BOOK-ONLY EDITION · Verified against the textbook

Selected from Chapter 20 · Chapter Summary in the textbook.

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STAT 3202 · Chapter revision

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Quick revision · 1

- A single factor experiment varies only one factor, the treatments, while all other conditions are kept uniform.
- CRD uses randomization and replication without blocking. It is most useful when units are relatively homogeneous.
- CRD is best suited to laboratory and greenhouse experiments; it is rarely used for field experiments because field plots are usually not homogeneous.
- RCBD adds local control (blocking) to randomization and replication, grouping heterogeneous experimental material into homogeneous blocks along a single known source of variation, most often a fertility gradient.
- In RCBD, number of blocks = number of replications, and number of plots per block = number of treatments.

Quick revision · 2

- LSD controls variation in two perpendicular directions at once, using rows and columns as two separate blocking factors, so the number of rows, columns, and treatments are all equal.
- The statistical model of CRD matches one-way ANOVA; the model of RCBD matches two-way ANOVA; LSD requires a three-way classification (rows, columns, treatments).
- An error-df target such as 12 is a practical planning guide, not a universal requirement. Replication should also reflect the desired power, precision, anticipated variability, and available resources.
- When the overall F test for treatments is significant, Fisher's least significant difference test is used to compare treatment pairs.
- Treatment means that share a common letter or symbol in a grouping table are not significantly different from each other.

Quick revision · 3

- Among CRD, RCBD, and LSD (for the same number of treatments and replications), CRD gives the largest error degrees of freedom, since none of the variation is set aside for blocking.
- Latin squares are often used for a modest number of treatments, for practical reasons rather than a mathematical restriction: too few treatments leave almost no error degrees of freedom, and too many require an impractically large number of homogeneous plots.

Important formulae · 1

Total number of experimental units in an equally replicated CRD:

$$n = rv$$

Error degrees of freedom in an equally replicated CRD:

$$edf = n - v = v(r-1)$$

Error degrees of freedom in RCBD:

$$edf = (v-1)(r-1)$$

Important formulae · 2

Error degrees of freedom in LSD:

$$edf = (v-1)(v-2)$$

Critical difference for CRD with equal replication:

$$C.D. = t_{\left(\frac{\alpha}{2},\, error\ df\right)}\sqrt{\frac{2\,MSE}{r}}$$

Critical difference for CRD with unequal replication:

$$C.D. = t_{\left(\frac{\alpha}{2},\, error\ df\right)}\sqrt{MSE\left(\frac{1}{r_i}+\frac{1}{r_j}\right)}$$

Important formulae · 3

Critical difference for RCBD:

$$CD = t_{\frac{\alpha}{2},edf}\sqrt{\frac{2\,MSE}{r}}$$

Critical difference for LSD:

$$C.D. = t_{\left(\frac{\alpha}{2},\, error\ df\right)}\sqrt{\frac{2\,MSE}{v}}$$

Correction factor for LSD:

$$C.F. = \frac{\left(\sum y_{ijk}\right)^2}{v^2}$$

Self-check from the chapter

1. The full form of CRD is Completely Randomized _____. [Design]
2. CRD is most suitable when experimental units are relatively _____. [Homogeneous]
3. The error degrees of freedom in a CRD with N observations and v treatments are _____. [$N - v$]
4. In an RCBD, each block contains every treatment exactly _____. [Once]